ENME 561/661 Lab 2

Fall 2026

Lab Safety

Emergency equipment (First aid kit, Fire extinguisher, wash stations) is located by each door

In case of emergency the closest exit is out the doors to the right and assemble in ICT

Lab PPE: closed-toed shoes. Tie back long hair and roll back loose sleeves before running the motor

In case of any near miss or incident report it to the TA and the Lab Tech

Lab Safety (continued)

- Come to the lab **on time**
- Place your coats and bags in the cubbies next to the door
- No Eating or Drinking in the lab
- Be careful when using the equipment, so you don't injure yourself or damage the equipment
  - Main concern is from rapidly moving parts
  - Keep the power off the equipment when connecting wires
  - Before using a device for the first time, read the user manual first.



Speed Friendship Activity

- Stand up and find someone you have not met.
- Talk for **one minute** — then switch partners.
- Ten rounds, about 12 minutes in total.
- Meet as many classmates as you can — at least 8.

Tell each person:

1. Your name and program.
2. Why mechatronics interests you.
3. The most interesting thing you know about mechatronics.

Before you sit down: form your team of 3 (or 2).

Team Formation

- Form a group of 3 (preferred) or 2.
- No groups larger than 3.
- The class should end up with no more than 9 groups total.
- Develop your team contract after this session, outside of the lab.



End-of-Session Shutdown Procedure

- At the end of each lab session, please ensure the following steps are completed:
  1. Make sure the QUARC controller is stopped.
  2. Turn off QUBE-Servo 2, and unplug it.
  3. Shut down the laptop.
  4. Place the laptop on the cabinet

Getting Started



LOG IN TO THE LAPTOP



ENSURE WIFI IS CONNECTED



RUN MATLAB 2022 B

AND OPEN SIMULINK

Hardware: QUBE-Servo 2

1



Encoder: *Digital E8P-512-118 single-ended optical shaft encoders. It outputs 2048 counts per revolution in quadrature mode (512 lines per revolution).*



2



3

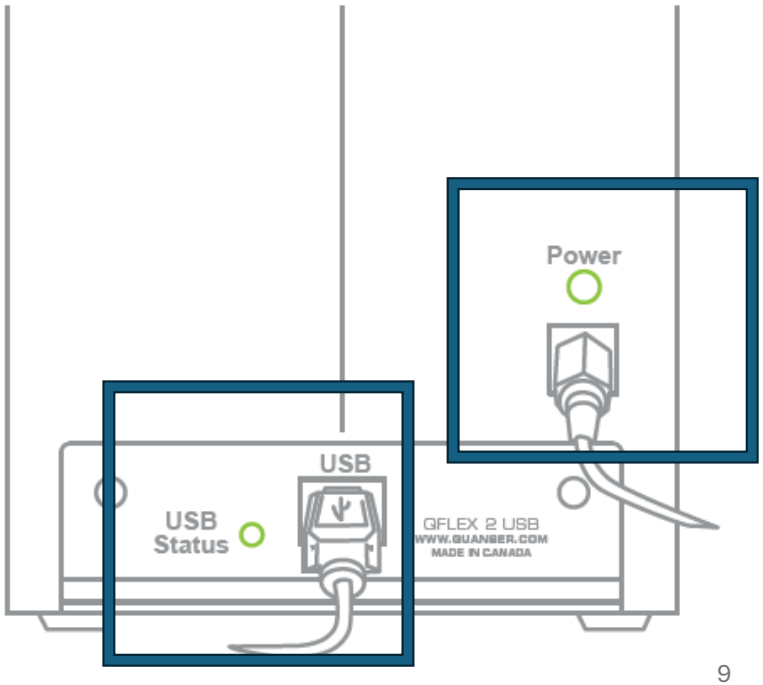


4

1. Quanser QUBE-Servo 2 (rotary DC motor and encoder)
2. USB 2.0 A/B cable
3. 24V, 2.71A power supply
4. Power cable
5. Laptop

Ensure that the same group member logs into the same laptop for every session and writes down the laptop number!

- First, connect the supplied 24V power supply to the Power connector on the QUBE-Servo 2 and to an outlet.
- Next, using the supplied USB cable, connect the QUBE-Servo 2 USB connector on the QFLEX 2 USB panel to an enabled USB 2.0 (or higher) port on your desktop PC or laptop.
- The Power LED on the QUBE-Servo 2 should light up green
- The USB Power LED on the QFLEX 2 USB panel should change from red to green.



Software

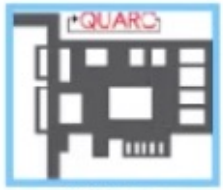
- MATLAB Simulink 2022 b or the latest installed on the lab computer
- QUARC library: It includes several custom blocks for interfacing with external hardware used in this class.

Configuring a Simulink Code for the QUBE-Servo 2

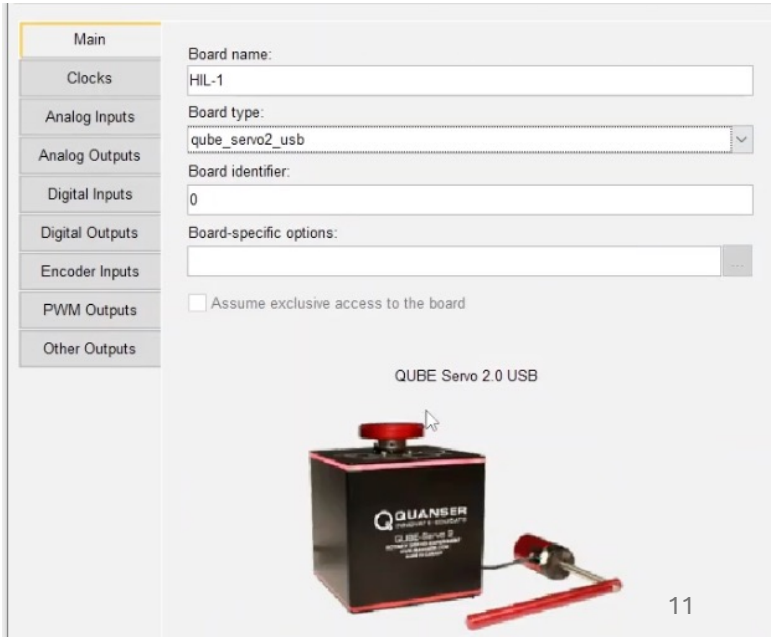
Open a Simulink blank model.
Write “HIL Initialize” to select this block Block.



Block

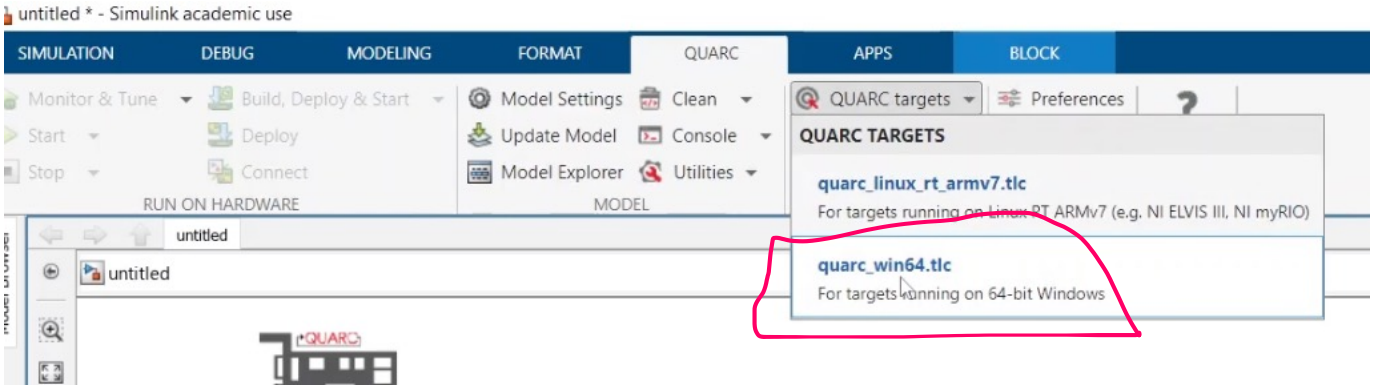


Then, double-click on the block, and choose the Board type to be “qube_sservo2_usb”.

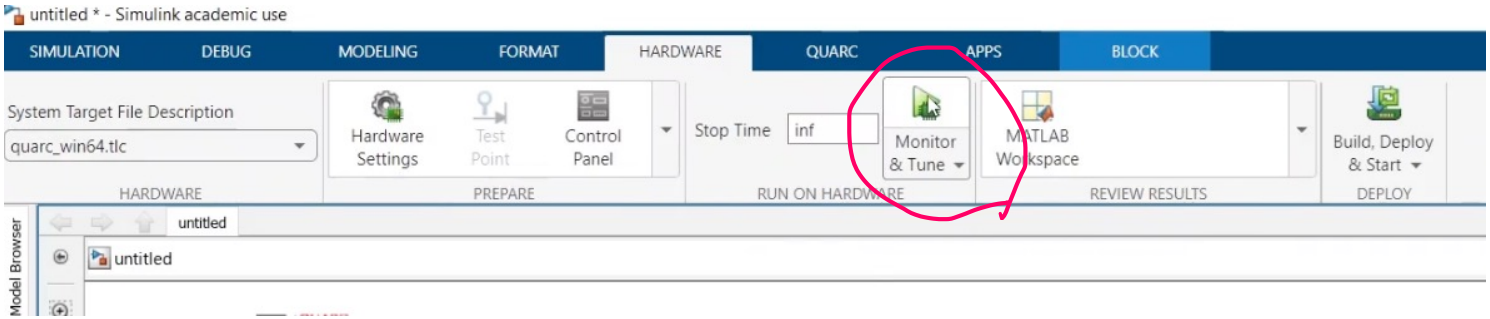


Running the program

From QUARC tab, select QUARC TARGETS, and select quarc_win64



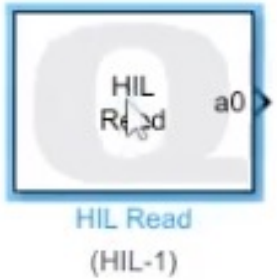
Then, Hardware tab appears, and then select Monitor &Tune



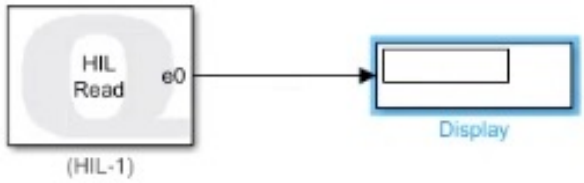
Then, the LED status light turns green

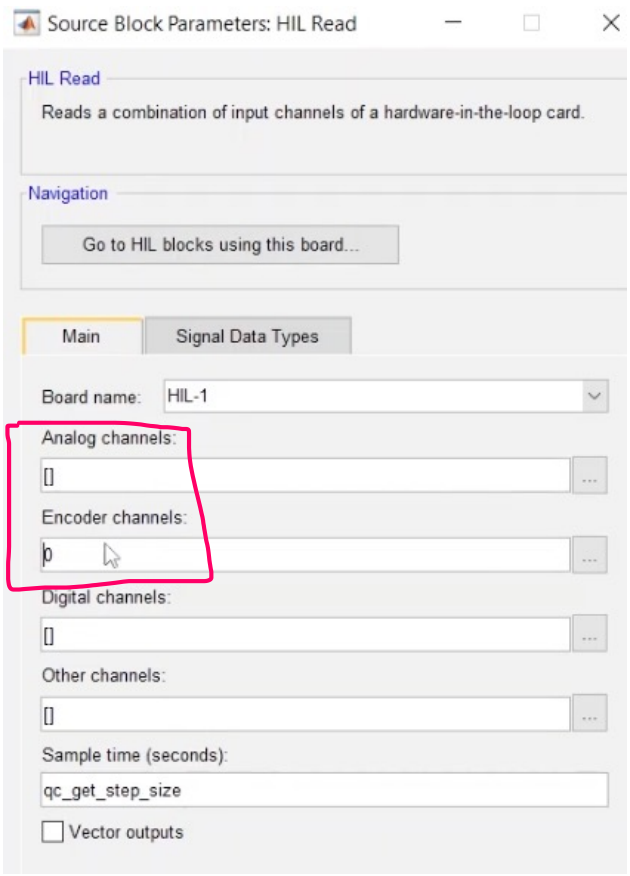
Reading Raw Encoder

- To read the raw encoder signal, type HIL Read, to add this block.
- Then, you need to read from the encoder channel 0.



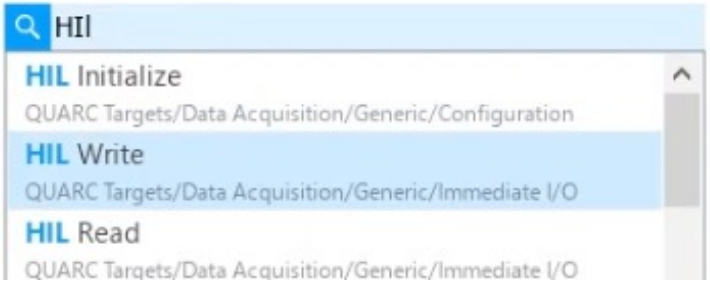
- Then connect it to a display block





Driving the DC Motor

- To send control commands to the motor type HIL Write, to add this block. This is to write to the analog port a0

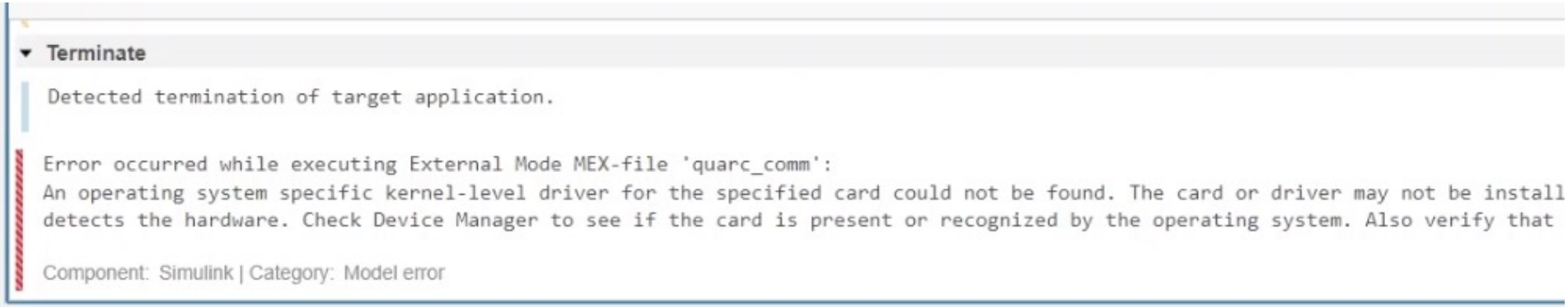




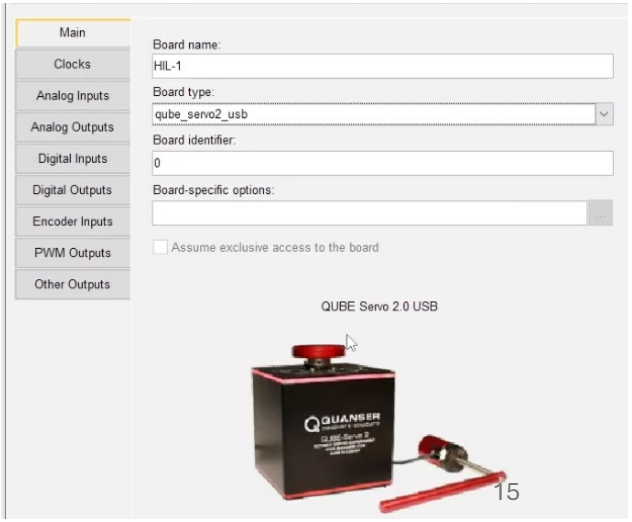
- Add the constant block
- Connect this block to the HIL



Common Error



- Ensure the USB is connected.
- Verify that the correct board is selected.
- If necessary, turn the device off and on by unplugging and reconnecting the power cable.



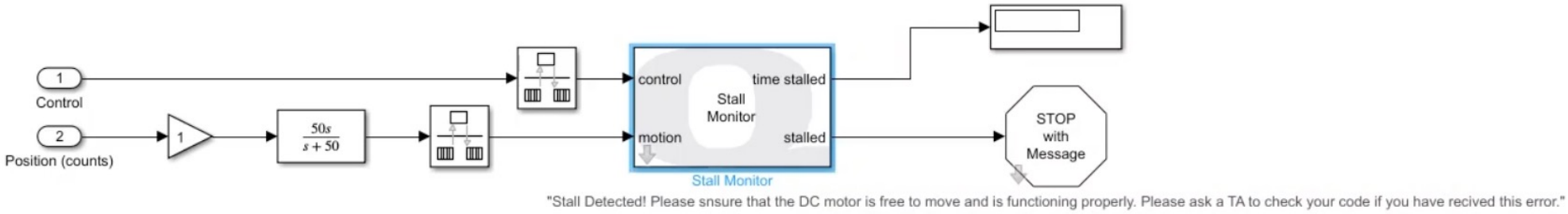


Caution

- Max motor input 10 V.
- Holding the motor in a stalled position for a prolonged period of time at applied voltages of over 5V can result in permanent damage.
- **When a lab asks you to stall the motor by hand:** reduce the voltage first, grip the edge of the disc rather than its face, hold only for the few seconds the reading needs, then release. Never grab the disc while it is running at full speed.

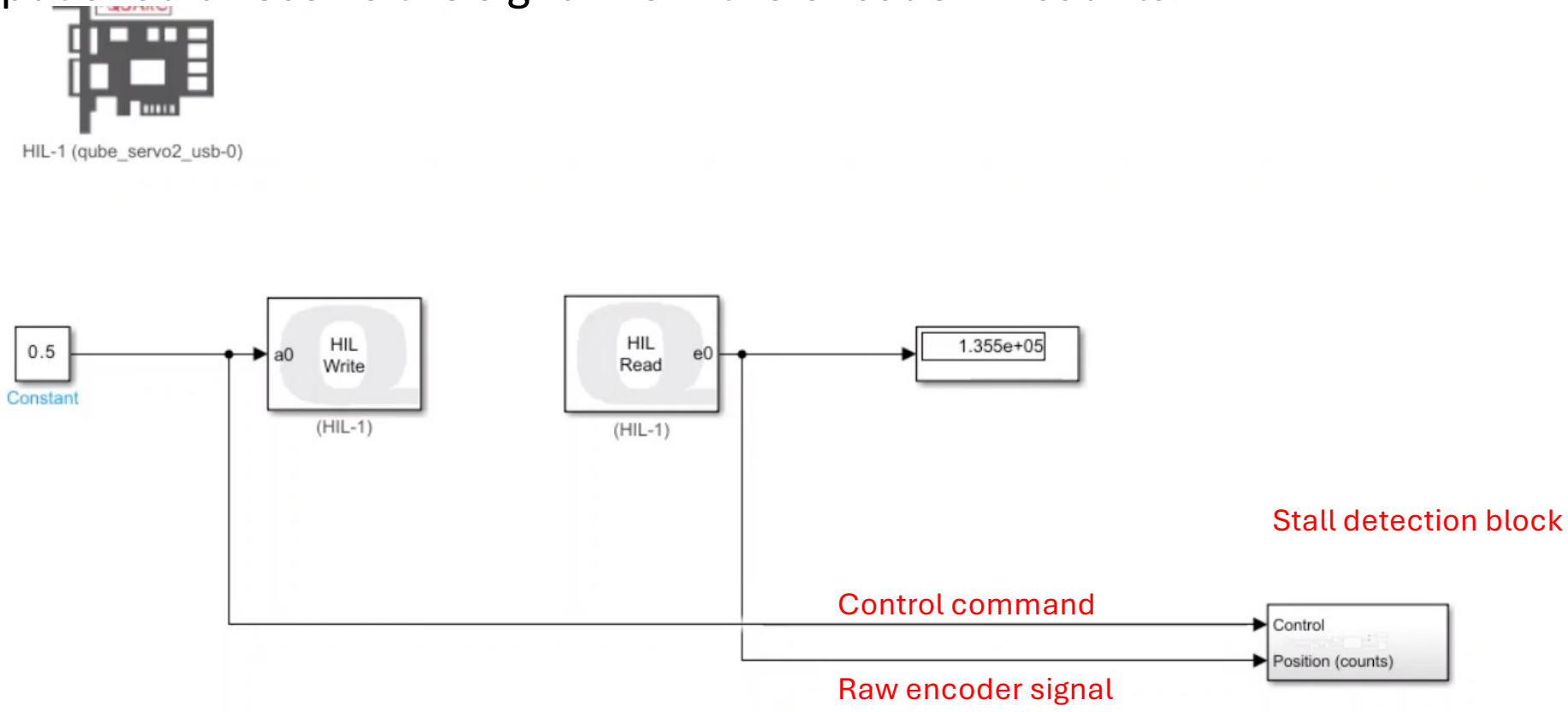
Adding the stall detection block

- It ensures the device is not receiving more than 5 volts or is not stalled.
- Download the file from D2L

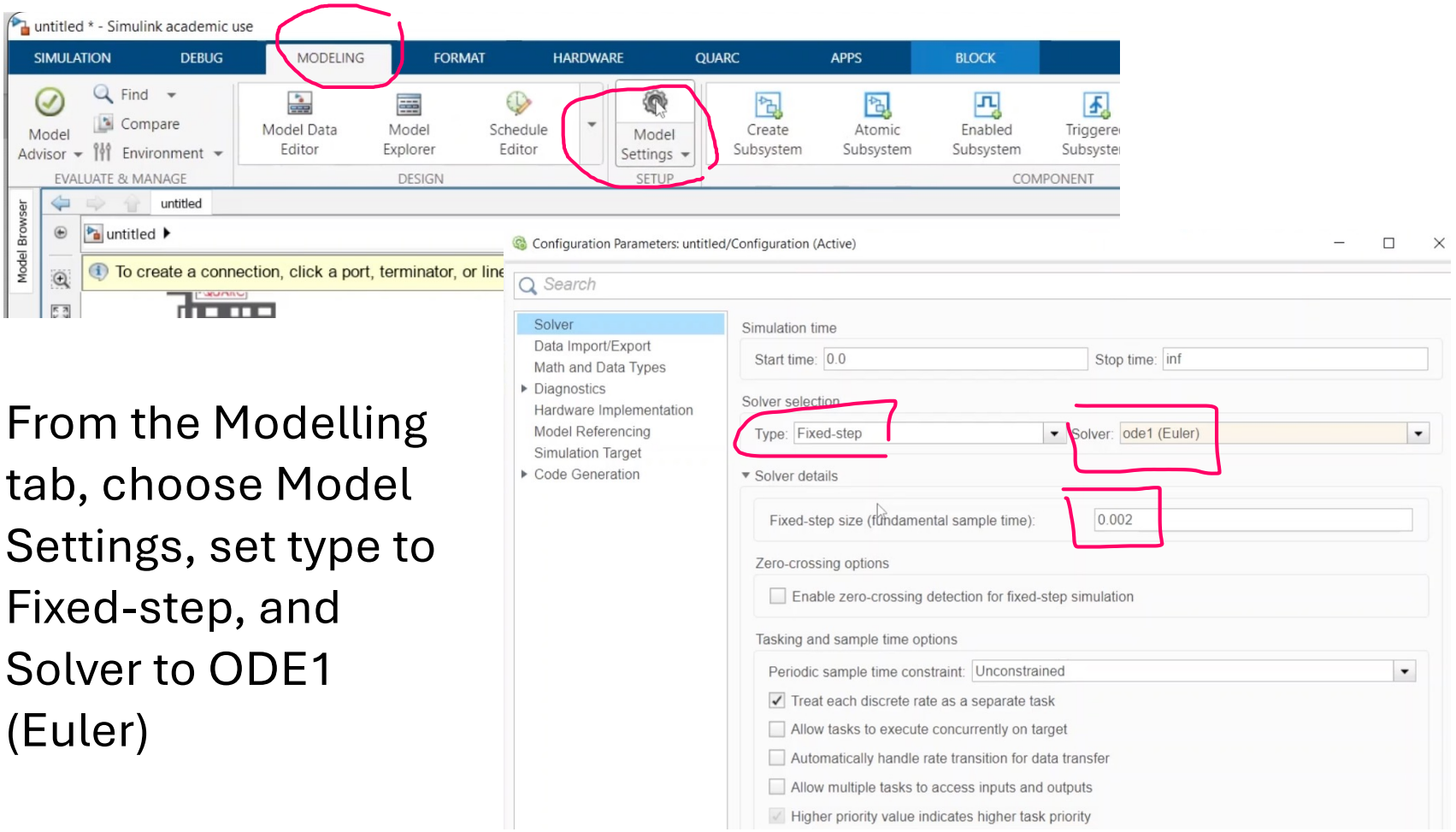


- Connect the Stall detection block to your system

The **Control** input for the stall detection block should be connected to the signal that you are sending to the HIL Write Block. The **Position (Counts)** input should receive the signal from the encoder in counts.



Solver



From the Modelling tab, choose Model Settings, set type to Fixed-step, and Solver to ODE1 (Euler)

Section 7: Design Your Own Experiment

- Section 7 is a **design task** — there are no steps to follow:
    1. Design an experiment to identify the motor resistance and the back-emf constant.
    2. Complete the requirements and constraints table (Table 2) **before** writing the procedure.
    3. Your procedure must determine the two parameters **independently** of one another.
    4. Submit it as a **separate PDF**, and bring a **printed copy to Lab 3**.
    5. In Lab 3 you will review another team's design, and they will review yours.
- **Marks: 70% your own design, 30% the quality of your review.**

Start the Lab

Open your **Lab Manual**

Follow the instructions step by step

Work with your **team (2–3 members)**

Ask questions if you get stuck

Section 6.5 needs no hardware — derive it and draft the .mlx at home.
Reuse your Forward Euler function from Lab 1.

References

- MATLAB Help Documentations
- Quanser MATLAB/Simulink and QUARC primer document

Report Guideline

- No formal report format required: you don’t need to list equipment or procedures.
- Focus on what you observed and the insights you gained.
- Address all questions in the lab handout.
- Questions marked “for your own understanding” need no written answer.
- Demonstrate your understanding of the lab material.
- Reports must be typed and submitted as a single PDF (separate images as solutions are not accepted)
  - Pages must be numbered
  - Figures and tables must have a caption
    - Table captions must be above tables, and figure captions below the figures
  - Equations must be numbered
- Upload all code files that you created to D2L, appropriately named
  - Simulink files (.slx format)
  - MATLAB file (.mlx format)
- Submit to D2L (do not compress files)
- No submission will be accepted by email
- **Section 7 is a separate PDF — bring a printed copy to Lab 3.**
- **Complete the Lab 2 exit survey on D2L before the deadline.**